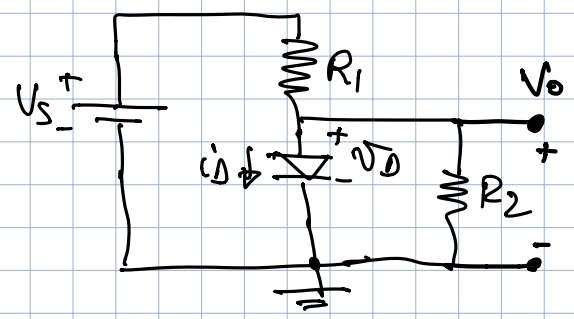
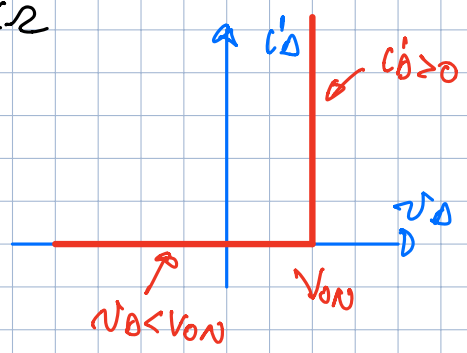
ESERCIZIO



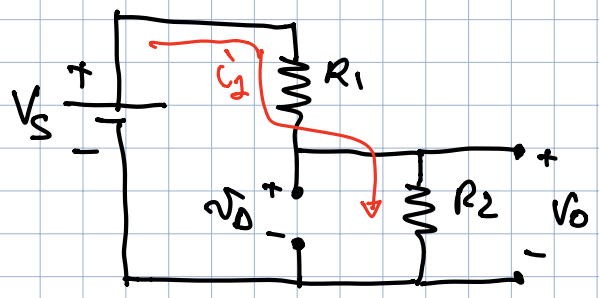
$V_S = 10V$

DIODO: $V_{ON} = 0{,}6V$ $R_{ON} = 0\Omega$

$R_1 = 2K\Omega$ $R_2 = 3K\Omega$



HP. DIODO = OFF ($v_D < V_{ON}$)

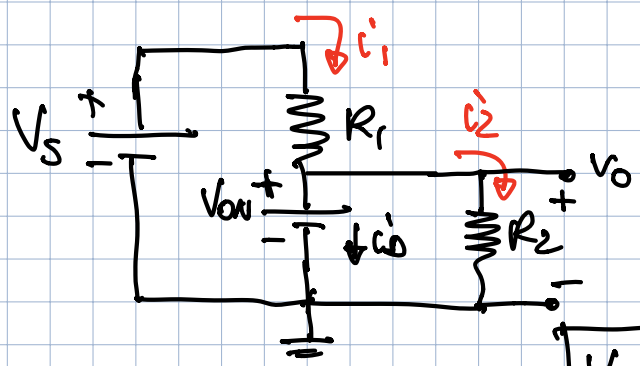


$$i_1 = \frac{V_S}{R_1+R_2} = \frac{10V}{5K\Omega} = 2mA$$

$$v_D = V_O = V_{R_2} = i_1 R_2 = 6V$$

$v_D > V_{ON} \Rightarrow$ DIODO NON È SPENTO MA "ON"

HP DIODO "ON" ($i_D \geq 0$)



$$i_2 = \frac{V_{ON}}{R_2} = \frac{0{,}6V}{3K\Omega} = 0{,}2mA$$

$$i_1 = \frac{V_S - V_{ON}}{R_1} = \frac{10V - 0{,}6V}{2K\Omega} = 4{,}7mA$$

$$i_D = i_1 - i_2 = 4{,}5mA$$

$i_D \geq 0$ OK DIODO ON

$$\boxed{V_O = V_{ON}}$$

AGGIUNGO DIFFICOLTÀ:

Se V_S VARIA, trovare $v_O = f(V_S)$ (TRANSCARATTERISTICA)

HP DIODO OFF (vedi SOPRA)

$$v_D = i_1 R_2 = V_S \cdot \frac{R_2}{R_1+R_2}$$

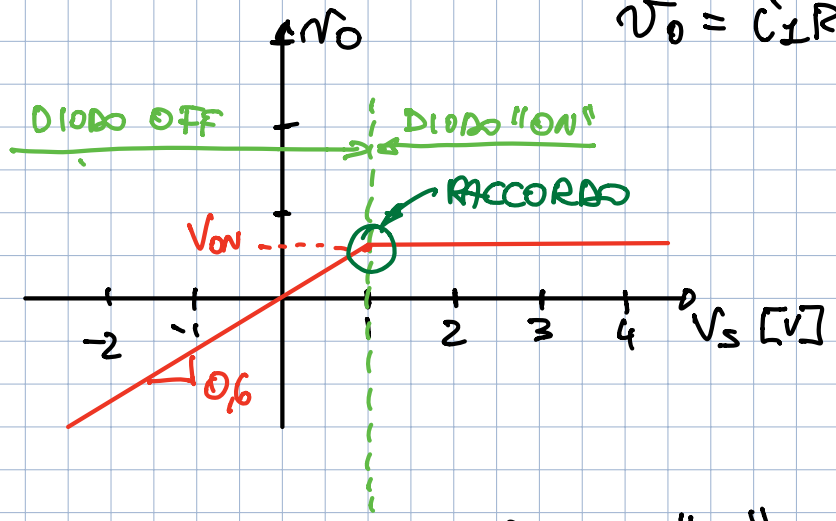
AFFINCHÉ DIOD OFF DEVE ESSERE $v_D < V_{ON}$

$$\Rightarrow V_S \frac{R_2}{R_1+R_2} < V_{ON} \Rightarrow V_S < V_{ON}\left(\frac{R_1+R_2}{R_2}\right)$$

$$\Rightarrow V_S < 0{,}6V \cdot \frac{5K\Omega}{3K\Omega} = 1V$$

QUINDI PER $V_S < 1V \Rightarrow$ DIODO = OFF e

$$v_0 = i_1 R_2 = V_S \cdot \frac{R_2}{R_1+R_2} = \frac{3}{5} V_S = 0{,}6\, V_S$$



PER $V_S = 1V \Rightarrow V_0 = 0{,}6V$

OK RACCORDO

PER $V_S > 1V \Rightarrow$ DIODO = "ON" $\Rightarrow v_0 = V_{ON} = 0{,}6V$

PER $V_S = 1V \Rightarrow V_0 = 0{,}6V$

VERIFICA "RIDONDANTE"

$$i_D = i_1 - i_2 = \frac{V_S - V_{ON}}{R_1} - \frac{V_{ON}}{R_2} > 0$$

$$\frac{V_S}{R_1} > \frac{V_{ON}}{R_1} + \frac{V_{ON}}{R_2}$$

$$\frac{V_S}{R_1} > V_{ON} \frac{R_1+R_2}{R_1 R_2}$$

$$V_S > V_{ON} \frac{R_1+R_2}{R_2} = 0{,}6V \cdot \frac{5}{3} = 1V$$

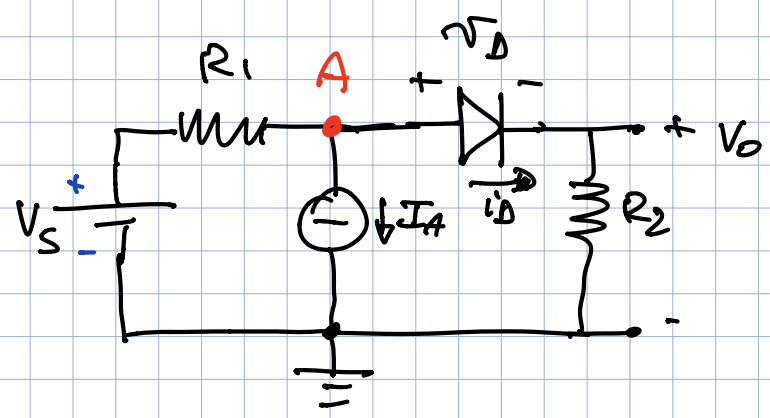
OK

PER $V_S > 1V \Rightarrow$ DIODO = ON e $V_0 = 0{,}6V$

IL DIODO "NON ZENER" HA DUE STATI POSSIBILI: ON e OFF; SE NON È IN UNO STATO NON PUÒ CHE ESSERE NELL'ALTRO!

È COMUNQUE UTILE CALCOLARE ANCHE LA CONDIZIONE x DIODO = ON IN QUANTO DEVE DARE LA STESSA SOGLIA! QUINDI È UN MODO x VERIFICARE IL RISULTATO

NELLA REALTÀ BISOGNA ANCHE VERIFICARE:
SE "ON" $i_D < I_{MAX}$
SE "OFF" $v_D > -V_{BREAK}$



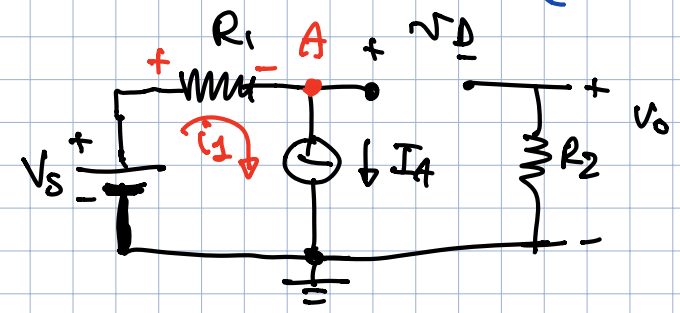
$R_1 = R_2 = 1\,K\Omega$

$I_A = 1\,mA$

$V_S = 6\,V$

$V_{ON} = 1\,V$

HP DIODO "OFF" ($v_D < V_{ON}$)

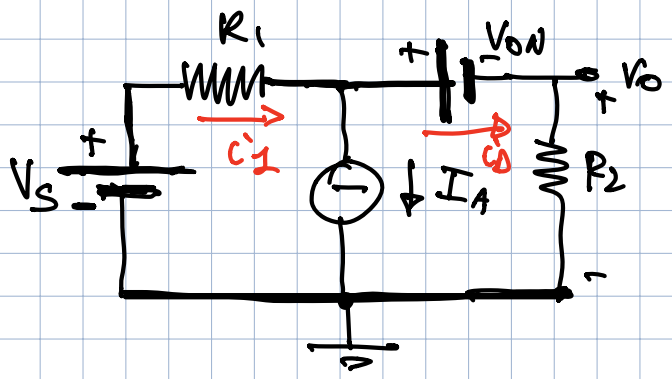


$V_A = V_S - R_1 I_A = 6\,V - 1\,mA \cdot 1\,K\Omega = 5\,V$

$v_D = V_A - V_O = 5\,V - 0 = 5\,V$

$v_D > V_{ON} \Rightarrow$ DIODO ON!

CONSIDERO DIODO "ON"



$V_S = R_1 i_1 + V_{ON} + R_2 i_D$

$i_D = i_1 - I_A$

$V_S = R_1 i_1 + V_{ON} + R_2 i_1 - R_2 I_A$

$\Rightarrow i_1 = \dfrac{V_S - V_{ON} + R_2 I_A}{R_1 + R_2} = \dfrac{6V - 1V + 1V}{2\,K\Omega} = 3\,mA$

$i_D = i_1 - I_A = 3\,mA - 1\,mA = 2\,mA$

$i_D > 0 \Rightarrow$ OK DIODO "ON"

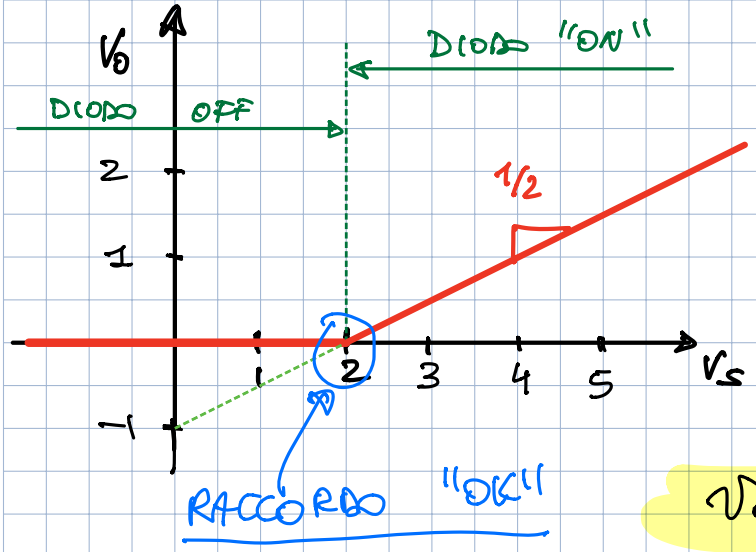
$v_O = i_D R_2 = 2\,V$

FACCIAMO V_S VARIABILE $\Rightarrow$ CONSIDERO DIODO "OFF"

$v_D = V_A = V_S - R_1 I_A < V_{ON}$

$\Rightarrow V_S < V_{ON} + R_1 I_A = 1\,V + 1\,V = 2\,V$!

CON $V_S < 2V$ DIODO = OFF e $V_O = 0$!



Con $V_S > 2V \Longrightarrow$ DIODO "ON"

$$i_D = i_1 - I_A = \frac{V_S - V_{ON} + R_2 I_A}{R_1 + R_2} - I_A$$

$$v_0 = R_2 i_D$$

$$= \frac{R_2}{R_1+R_2} V_S + \frac{R_2}{R_1+R_2}\underbrace{(-V_{ON} + R_2 I_A)}_{-1+1=0} - I_A R_2$$

$$v_0 = \frac{1}{2} V_S - 1\ V \qquad (V_S > 2V)$$

$v_0(2V) = 0V$ OK RACCORDO